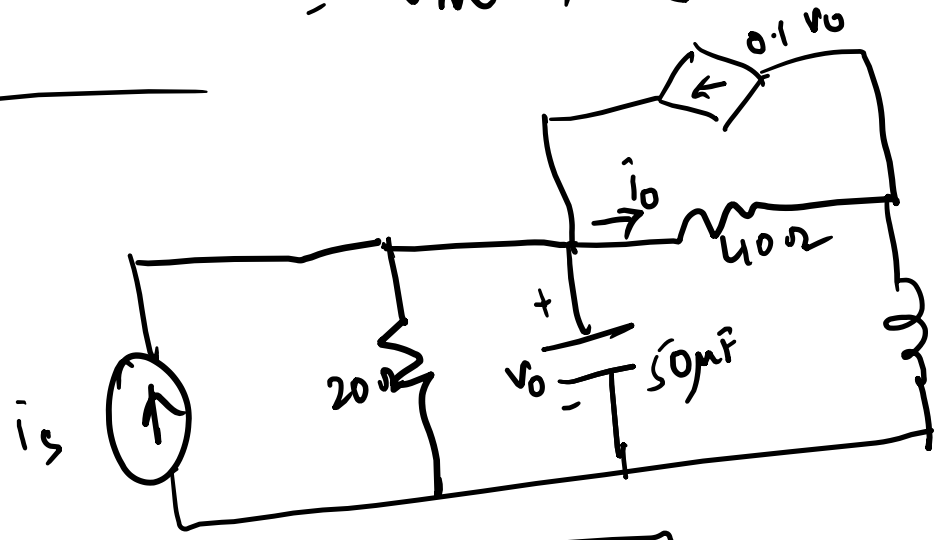
$$v(t) = V\sin(\omega t \pm \phi)$$

Phasor: $X = X_m \angle\theta$

$$v(t) = V_m \cos(\omega t + \phi)$$
$$= V_m \angle\phi$$
$$= V_m e^{j\phi}$$
$$= V_m\cos\phi + j V_m \sin\phi$$



Find i_o using nodal analysis. given

$i_s = 6\cos(200t + 15°)$ A

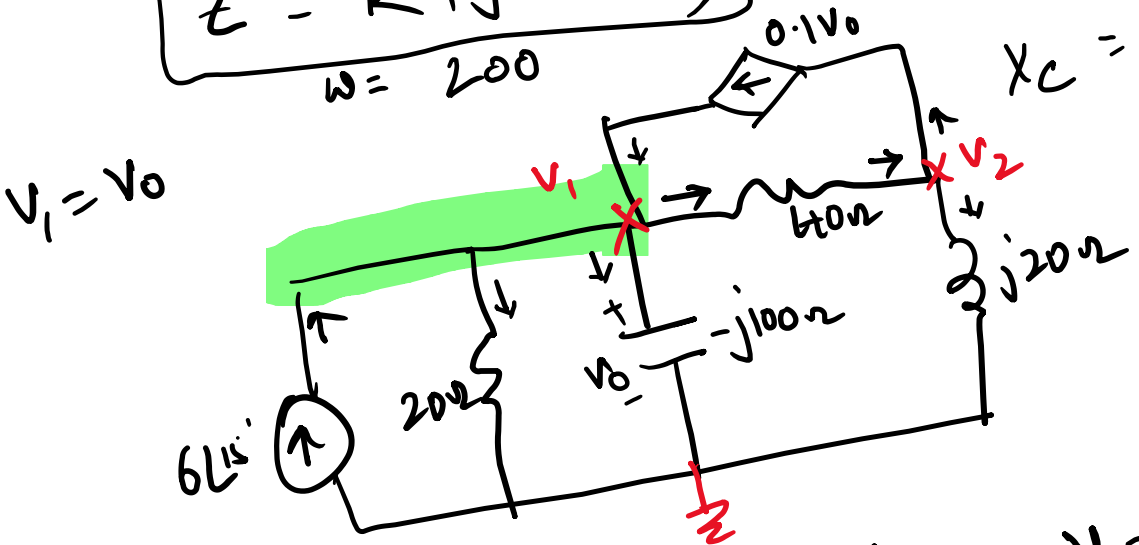
$= 6\angle 15°$

$X_L = 20\Omega$

$X_C = 100\Omega$

$$Z = R + j(X_L - X_C)$$

$\omega = 200$



$v_1 = v_o$

Node v_1: $6\angle 15° + 0.1v_1 = \frac{v_1}{20} + \frac{v_1}{-j100} + \frac{v_1 - v_2}{40}$

Node v_2: $\frac{v_1 - v_2}{40} = \frac{v_2}{j20} + 0.1v_o$

$X = X_m \cos(\omega t + \phi)$ → $(X_m \angle \phi)$ Polar

$(X_m \cos\phi) + j(X_m \sin\phi)$ Im.

Re Rectangular.

$$\frac{R_1 \angle \theta_1}{R_2 \angle \theta_2} = \frac{R_1}{R_2} \angle \theta_1 - \theta_2$$

$$R_1 \angle \theta_1 \times R_2 \angle \theta_2 = R_1 R_2 \angle \theta_1 + \theta_2$$

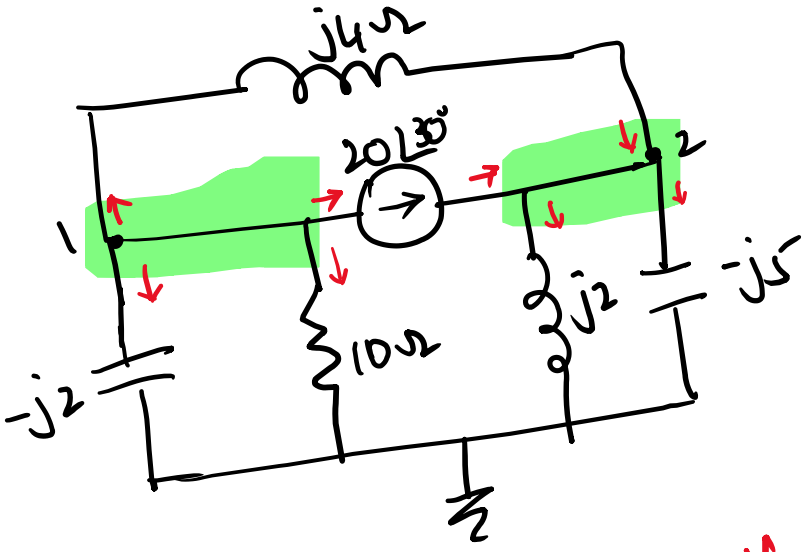
$$5.7955 + j\,1.5529 = V_1(-0.025 + j0.01) - 0.025 V_2 \quad -(1)$$

$$0 = 3V_1 + (1 - j2)V_2 \quad -(2)$$

$$V_1 = -70.63 - j127.33\ V$$

$$V_2 = -110.3 + j\,161.09\ V$$

$$i_o = \frac{V_1 - V_2}{40} = 7.276 \angle -82.17^\circ$$



Find node voltages.

Node 1: $\frac{V_1}{-j2} + \frac{V_1}{10} + \frac{V_1 - V_2}{j4} + 20\angle 30^\circ = 0$

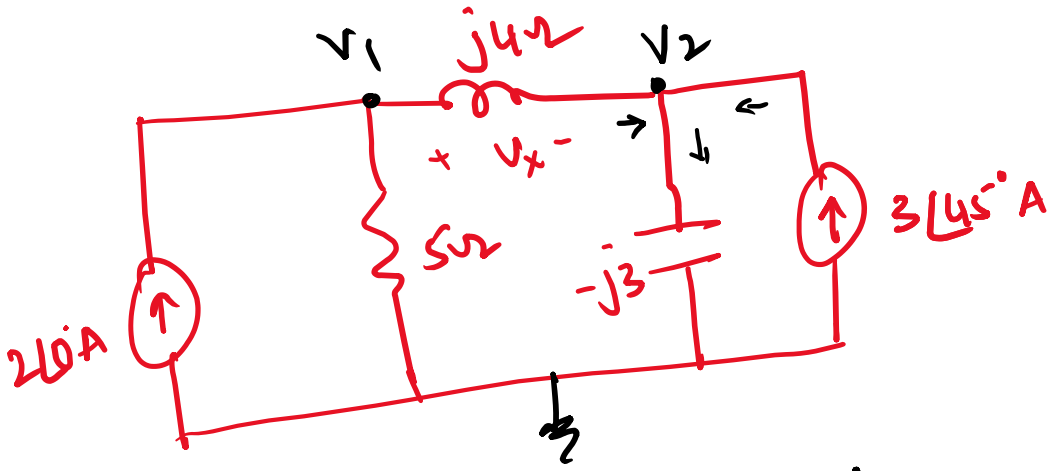
Node 2: $20\angle 30 + \frac{V_1 - V_2}{j4} = \frac{V_2}{j2} + \frac{V_2}{-j5}$

$$173.2 + j100 = -V_1(1 + j2.5) - j2.5V_2 \quad -①$$

$$173.2 + j100 = j2.5V_1 - j5.5V_2 \quad -②$$

$$V_1 = 28.93\angle 135.38^\circ$$

$$V_2 = 49.18\angle 124.08^\circ$$



Find V_x using nodal analysis

$$V_x = V_1 - V_2$$

At V_1: $2\angle 0 = \frac{V_1}{5} + \frac{V_1 - V_2}{j4}$

At V_2: $\frac{V_1 - V_2}{j4} + 3\angle 45 = \frac{V_2}{-j3}$

$$aV_1 + bV_2 = c$$